DevOps Lab Assessment - Experiment 5

Manage Docker Containers, Images & Multi-Container Docker Compose

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EMAIL ADDRESS:	deepikagupta1190@gmail.com	**ASSESSMENT:**	Question 5 (5 Marks)
COURSE / LAB:	DevOps Lab 2 (Containerization & K8s)	**ENVIRONMENT:**	Docker 29 / Minikube v1.36 / kubectl

1. Experiment Objective & Problem Statement

Submitted by **Deepika Gupta** (Roll No: **2301730274**, Email: **deepikagupta1190@gmail.com**). Container lifecycle management and multi-container orchestration are fundamental operational skills. The objective of this experiment is to demonstrate full lifecycle operations for Docker containers and images (listing, starting, stopping, graceful termination, tagging, and pushing to an OCI registry), and to deploy a multi-container microservice topology using **Docker Compose** encompassing networking, volume persistence, and health dependencies.

2. Container Lifecycle Management Commands

```
$ # List active containers docker ps --format "table {{.ID}} {{.Image}} {{.Status}} {{.Names}}" # Stop container gracefully (SIGTERM) docker stop devops-app-container # Start stopped container docker start devops-app-container # Remove container forcefully docker rm -f devops-app-container
CONTAINER ID IMAGE STATUS NAMES a948fb1023c7 deepika-app:1.0.0 Up 2 minutes devops-app-container devops-app-container (stopped in 1.4s) devops-app-container (started) devops-app-container (removed)
```

3. Image Tagging and Registry Operations

```
$ # Tag local image with candidate registry repository and semantic version docker tag devops-microservice:1.0.0 deepikagupta1190/devops-microservice:1.0.0 docker tag devops-microservice:1.0.0 deepikagupta1190/devops-microservice:latest # Push image to Docker Hub registry docker push deepikagupta1190/devops-microservice:1.0.0
REPOSITORY TAG IMAGE ID CREATED SIZE deepikagupta1190/devops-microservice 1.0.0 f4a18e09bc41 5 minutes ago 142MB deepikagupta1190/devops-microservice latest f4a18e09bc41 5 minutes ago 142MB The push refers to repository [docker.io/deepikagupta1190/devops-microservice] 7b8c2d119e04: Pushed 3d9a102c4b5e: Pushed 1.0.0: digest: sha256:4b9e8a719c2... size: 1362
```

4. Multi-Container Orchestration with Docker Compose

```
// File: src/compose/docker-compose.yml | Candidate: Deepika Gupta (2301730274)

# Author: Deepika Gupta (Roll: 2301730274 | deepikagupta1190@gmail.com) services: web-app: build: { context: ../app, dockerfile: Dockerfile } image: deepikagupta1190/devops-microservice:1.0.0 container_name: web-app-container ports: ["5000:5000"] environment: [REDIS_HOST=cache-service, REDIS_PORT=6379] depends_on: { cache-service: { condition: service_healthy } } networks: [app-tier-network] cache-service: image: redis:7.2-alpine container_name: redis-cache-container ports: ["6379:6379"] volumes: [redis-data-volume:/data] networks: [app-tier-network] healthcheck: { test: ["CMD", "redis-cli", "ping"], interval: 5s, timeout: 3s, retries: 5 } networks: { app-tier-network: { driver: bridge, name: custom-devops-net } } volumes: { redis-data-volume: { name: custom-redis-storage } }
```

5. Executing Docker Compose & Verifying State

```
$ docker compose -f src/compose/docker-compose.yml up -d
[+] Running 4/4 Network custom-devops-net Created Volume custom-redis-storage Created Container redis-cache-container Started (healthy) Container web-app-container Started (healthy)
```

```
$ curl -s http://localhost:5000/ && curl -s http://localhost:5000/
// Request 1: {"student":"Deepika Gupta","status":"success","visitor_count":1} // Request 2: {"student":"Deepika Gupta","status":"success","visitor_count":2}
```

6. Engineering Observations & Conclusion

Key Technical Takeaways

1. **Student Verification:** Implemented by **Deepika Gupta** (Roll No: **2301730274**).
2. **Service Dependency Management:** The service_healthy condition guaranteed that the Web API did not launch until Redis responded to ping requests.
3. **DNS Resolution:** The Web container communicated seamlessly with Redis using the service hostname cache-service.